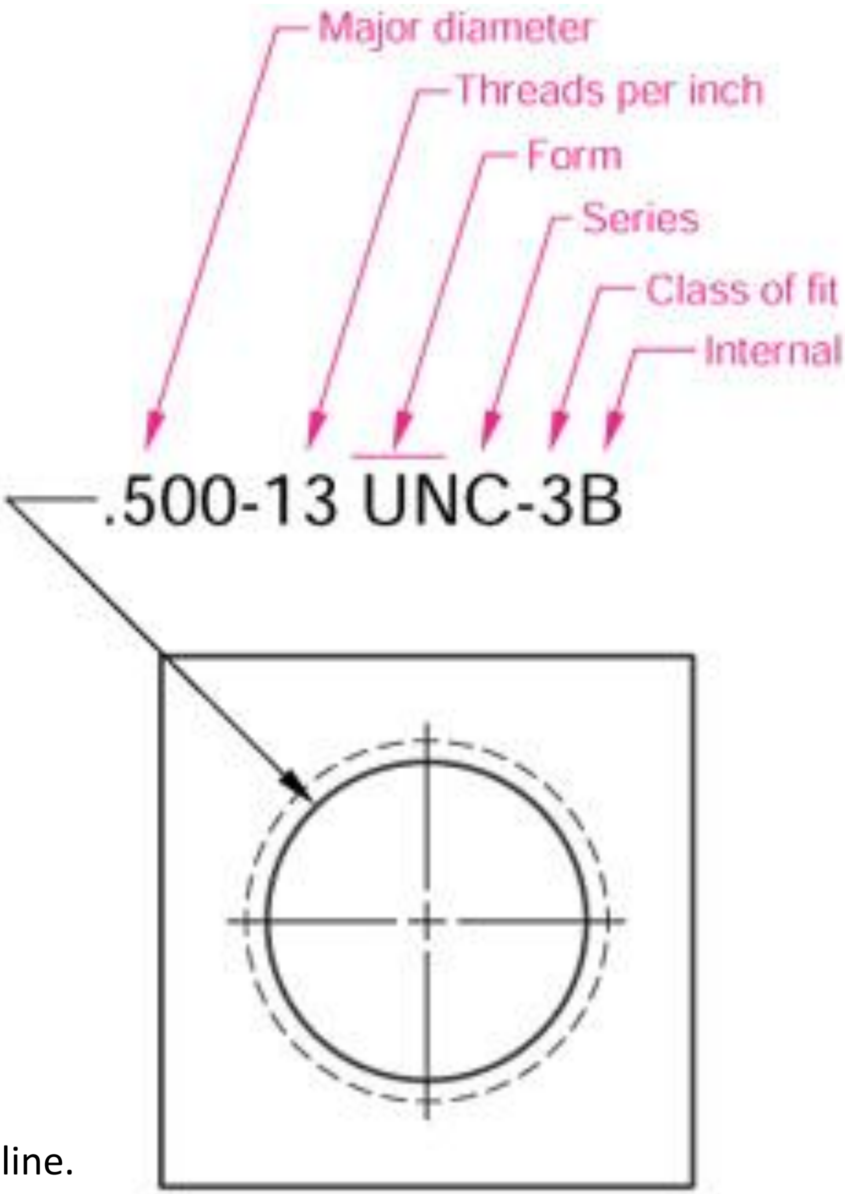
Fasteners

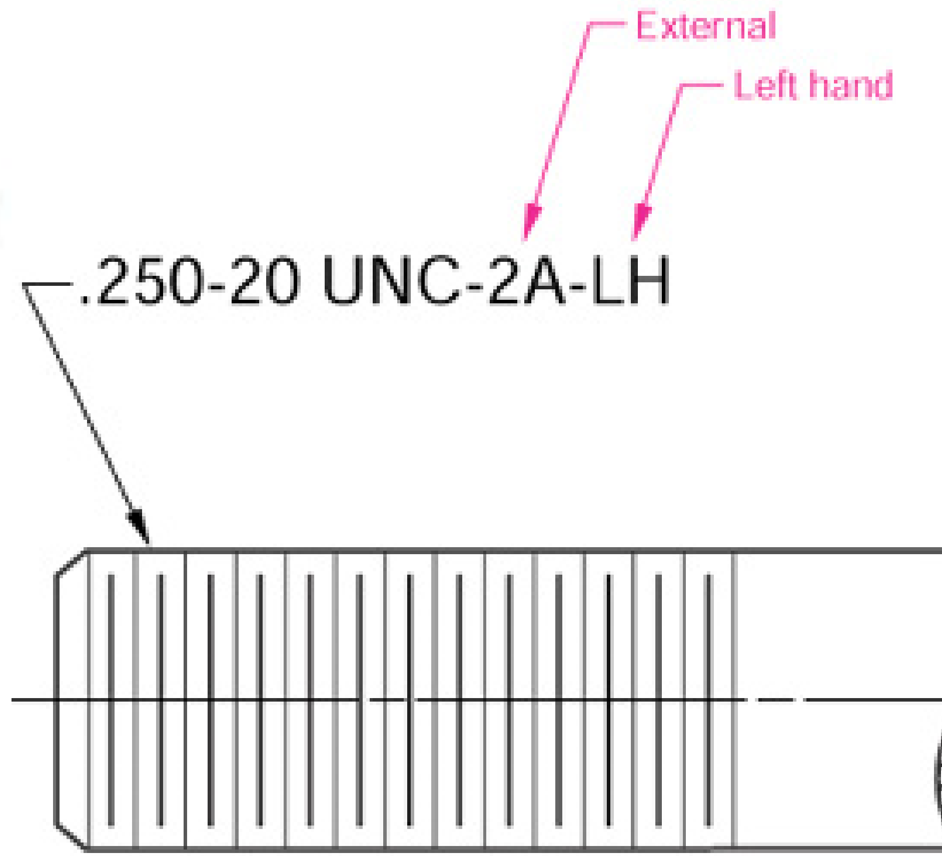
ME 350 Lecture

Mike Umbriac

Image: london2012.com

Thread Designations (Inch System)



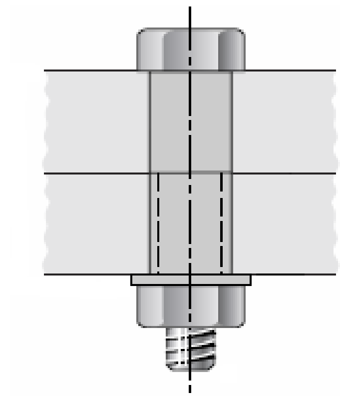


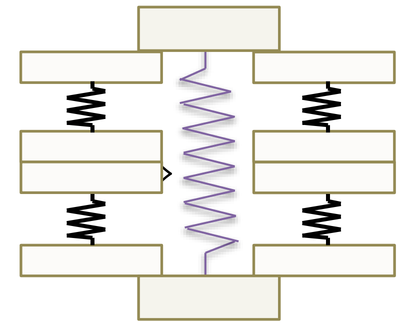
Bertoline.

Preload, F_i

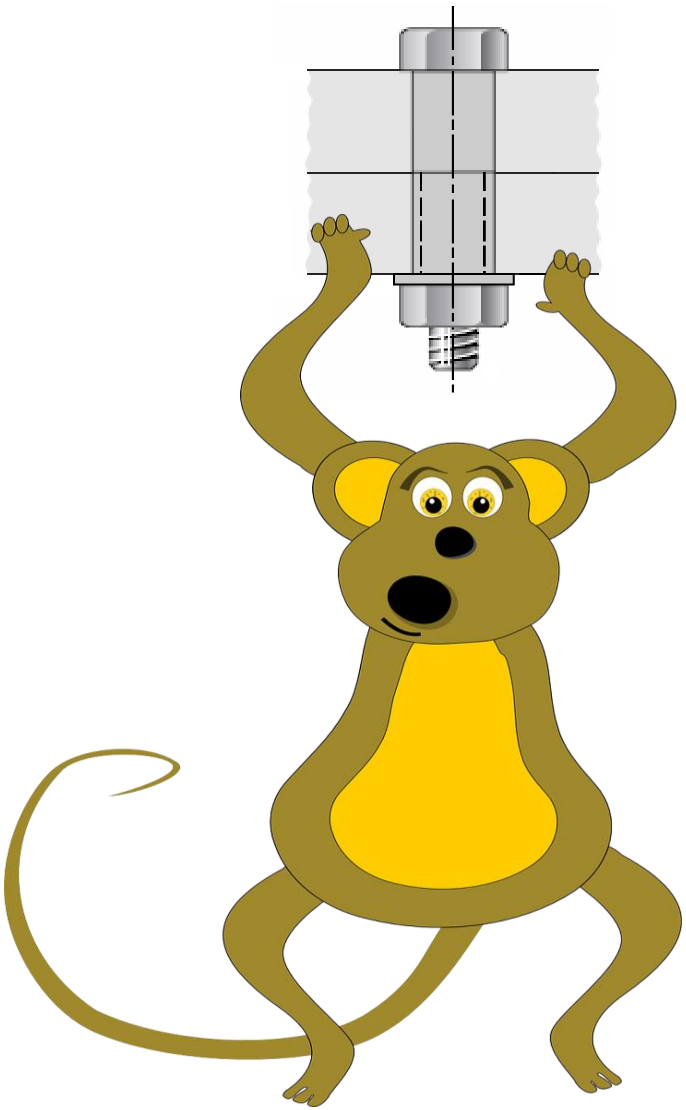
Tightening the joint elongates the bolt (and compresses the clamped members.)

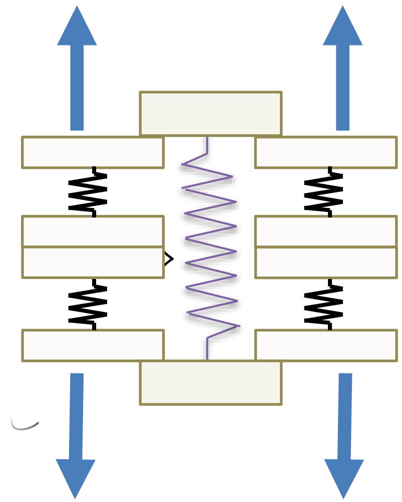
The axial tensile load in the bolt, that comes from the initial tightening of the bolt, is called the preload.





External load, *P*





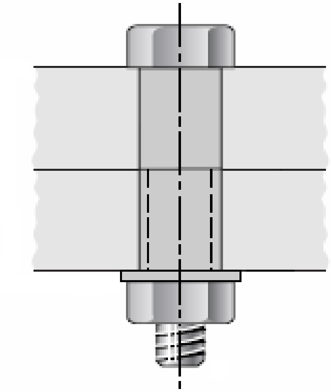
For our purposes, the external load is an applied load that wants to pull the members apart from each other.

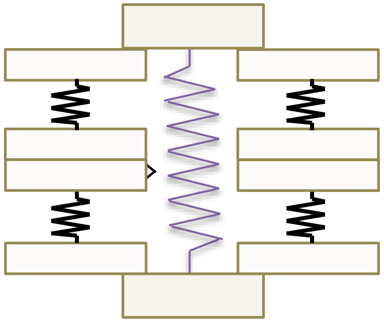
Does all of this load go into increasing the load on the bolt?

Image: https://www.needpix.com/photo/93375/monkey-ape-animal-hanging-swinging-zoo-funny-tail-yellow (CC)

Joint Constant, *C*

Tightening the joint elongates the bolt and compresses the clamped members.

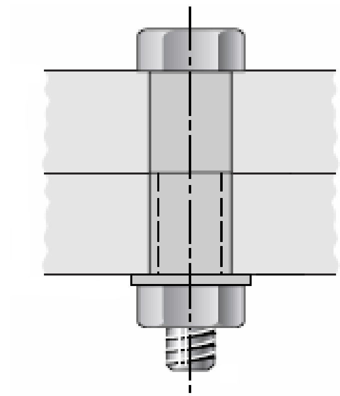


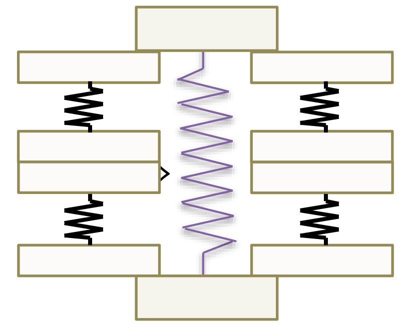


$$C = \frac{k_b}{k_b + k_m}$$

- k_b is the stiffness of the bolt
- k_m is the stiffness of the members being clamped

F_b, Force in the bolt, is calculated using the Joint Constant, C





As long as the joint does not separate,

$$F_b = \frac{k_b}{k_b + km} P + Fi$$

Joint Constant, C

- k_b is the stiffness of the bolt
- k_m is the stiffness of the members being clamped
- P is the external load on the joint
- F_i is the preload that comes from initially tightening the bolt

Tensile Strength of a Threaded Fastener

Table 8-2

Tensile Stress Area, A_t

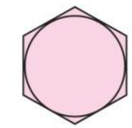
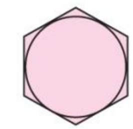
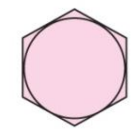
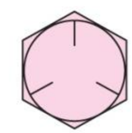
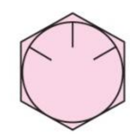
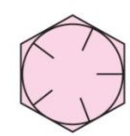
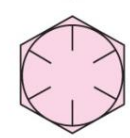
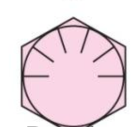
Size Designation	Nominal Major Diameter in	Coarse Series—UNC Threads per Inch N	Coarse Series—UNC Tensile-Stress Area A_t in²	Coarse Series—UNC Minor-Diameter Area A_r in²	Fine Series—UNF Threads per Inch N	Fine Series—UNF Tensile-Stress Area A_t in²	Fine Series—UNF Minor-Diameter Area A_r in²
0	0.0600				80	0.001 80	0.001 51
1	0.0730	64	0.002 63	0.002 18	72	0.002 78	0.002 37
2	0.0860	56	0.003 70	0.003 10	64	0.003 94	0.003 39
3	0.0990	48	0.004 87	0.004 06	56	0.005 23	0.004 51
4	0.1120	40	0.006 04	0.004 96	48	0.006 61	0.005 66
5	0.1250	40	0.007 96	0.006 72	44	0.008 80	0.007 16
6	0.1380	32	0.009 09	0.007 45	40	0.010 15	0.008 74
8	0.1640	32	0.014 0	0.011 96	36	0.014 74	0.012 85
10	0.1900	24	0.017 5	0.014 50	32	0.020 0	0.017 5
12	0.2160	24	0.024 2	0.020 6	28	0.025 8	0.022 6
$\frac{1}{4}$	0.2500	20	0.031 8	0.026 9	28	0.036 4	0.032 6
$\frac{5}{16}$	0.3125	18	0.052 4	0.045 4	24	0.058 0	0.052 4
$\frac{3}{8}$	0.3750	16	0.077 5	0.067 8	24	0.087 8	0.080 9
$\frac{7}{16}$	0.4375	14	0.106 3	0.093 3	20	0.118 7	0.109 0
$\frac{1}{2}$	0.5000	13	0.141 9	0.125 7	20	0.159 9	0.148 6
$\frac{9}{16}$	0.5625	12	0.182	0.162	18	0.203	0.189
$\frac{5}{8}$	0.6250	11	0.226	0.202	18	0.256	0.240
$\frac{3}{4}$	0.7500	10	0.334	0.302	16	0.373	0.351
$\frac{7}{8}$	0.8750	9	0.462	0.419	14	0.509	0.480
1	1.0000	8	0.606	0.551	12	0.663	0.625
$1\frac{1}{4}$	1.2500	7	0.969	0.890	12	1.073	1.024
$1\frac{1}{2}$	1.5000	6	1.405	1.294	12	1.581	1.521

Proof Strength, S_p

- The proof load is the maximum load a bolt can withstand without acquiring a permanent set.
- The proof strength S_p is the proof load divided by the tensile stress area A_t.
- The bolt grades are numbered according to tensile strength. (SAE grade 5, grade 8, etc.) This is sometimes called a "property class."

Proof Strength S_p and Tensile Strength S_{ut}

Table 8–9

SAE Grade No.	Size Range Inclusive, in	Minimum Proof Strength,* kpsi	Minimum Tensile Strength,* kpsi	Minimum Yield Strength,* kpsi	Material	Head Marking
1	$\frac{1}{4}$–$1\frac{1}{2}$	33	60	36	Low or medium carbon	
2	$\frac{1}{4}$–$\frac{3}{4}$	55	74	57	Low or medium carbon	
	$\frac{7}{8}$–$1\frac{1}{2}$	33	60	36		
4	$\frac{1}{4}$–$1\frac{1}{2}$	65	115	100	Medium carbon, cold-drawn	
5	$\frac{1}{4}$–1	85	120	92	Medium carbon, Q&T	
	$1\frac{1}{8}$–$1\frac{1}{2}$	74	105	81		
5.2	$\frac{1}{4}$–1	85	120	92	Low-carbon martensite, Q&T	
7	$\frac{1}{4}$–$1\frac{1}{2}$	105	133	115	Medium-carbon alloy, Q&T	
8	$\frac{1}{4}$–$1\frac{1}{2}$	120	150	130	Medium-carbon alloy, Q&T	
8.2	$\frac{1}{4}$–1	120	150	130	Low-carbon martensite, Q&T	

Recommended Preload, F_i

- The preload F_i is the clamping force produced by the bolt.
- It helps keep the bolt or nut from coming loose, and also helps keep the joint from separating under external load.

$$F_i = 0.75 S_p A_t \text{ for nonpermanent connections, reused fasteners}$$

$$F_i = 0.9 S_p A_t \text{ for permanent connections}$$

Calculating the Target Torque T based on the Preload F_i

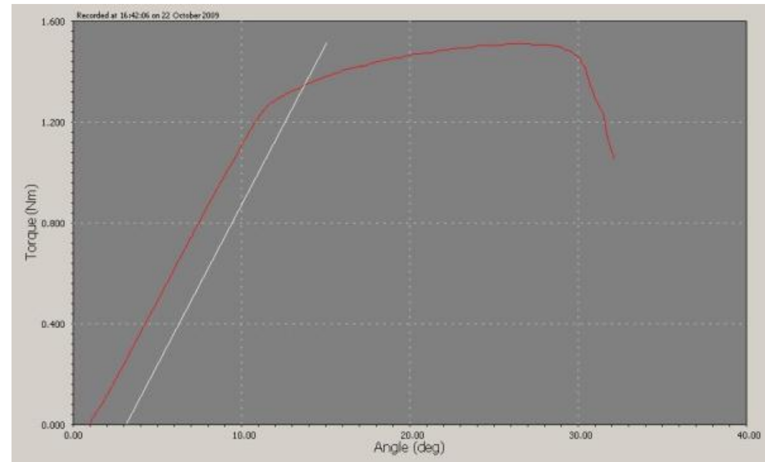
$$T = K F_i d$$

K = .2 for most applications

d = major diameter of the fastener

For more detail, see *Shigley's Mechanical Engineering Design* textbook

Nutrunners are often used in factories to ensure the correct torque is achieved.



Torque-angle curve

Images: directindustry.com, info.admet.com, ferret.com.au

STRETCH!

Safety Factors

- We always analyze an approximation of reality, never reality itself.
- This uncertainty is expressed traditionally with the Safety Factor (SF): Multiply a calculated (estimated) load by a number that reflects this uncertainty, e.g.,

  Estimated Stress x SF = Material Strength

  Here SF represents uncertainty in both stress and strength estimation
- The SF also reflects the consequences of failure and the cost of overdesign: Reflects risk.
- Some “typical” SFs: Buildings and Machines SF= 2.0, pressure vessels SF = 3.5-4.0, aircraft SF= 1.5-2.0

Factor of Safety against Joint Separation, n_s

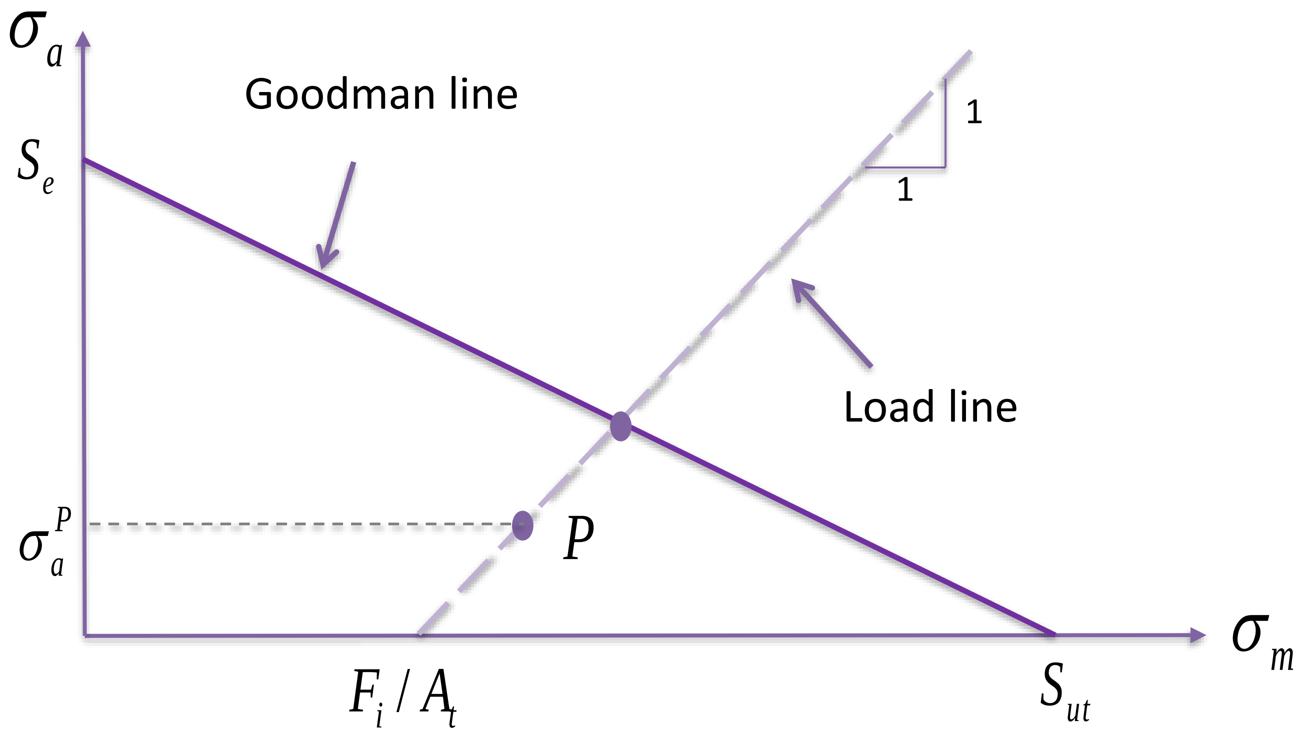
This factor prevents the joint from separating, which would cause the bolt to take all the external load.

$$n_s = \frac{F_i}{(1-C)P}$$

(F_i is the preload, C is the joint constant, and ***P* is the external load**.)

Factor of Safety Against Fatigue Failure, n_f

Using the Modified Goodman Line, and for external loads that fluctuate between 0 and a maximum force P



$$n_f = \frac{S_{ut} A_t - F_i}{1 + S_{ut} / S_e} \times \frac{2}{CP}$$

Source: Shigley; Image: Saitou

Endurance Strength, S_e

Table 8–17

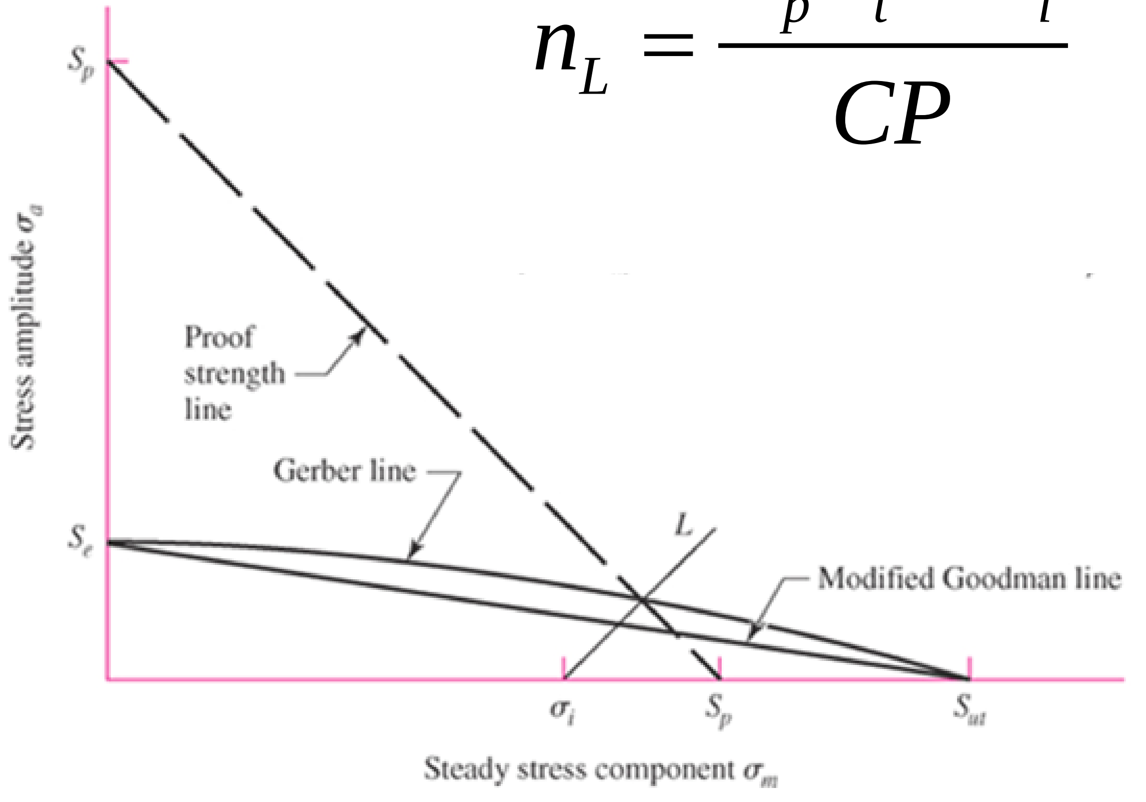
Fully Corrected Endurance Strengths for Bolts and Screws with Rolled Threads*

Grade or Class	Size Range	Endurance Strength
SAE 5	$\frac{1}{4}$–1 in	18.6 kpsi
	$1\frac{1}{8}$–$1\frac{1}{2}$ in	16.3 kpsi
SAE 7	$\frac{1}{4}$–$1\frac{1}{2}$ in	20.6 kpsi
SAE 8	$\frac{1}{4}$–$1\frac{1}{2}$ in	23.2 kpsi
ISO 8.8	M16–M36	129 MPa
ISO 9.8	M1.6–M16	140 MPa
ISO 10.9	M5–M36	162 MPa
ISO 12.9	M1.6–M36	190 MPa

*Repeatedly applied, axial loading, fully corrected.

Load Factor against Bolt Yielding, n_L

$$n_L = \frac{S_p A_t - F_i}{CP}$$



Example Problem: Fasteners in Fatigue

A bolted joint is subjected to a fluctuating tensile load whose maximum value is 4 kip (kilopounds). The bolt is a ½-13 UNC thread, SAE Grade 8. Use a value of 5.1 Mlbf/in for k_b (stiffness of the bolt), and 15.4 Mlbf/in for k_m (stiffness of the members). Assume that the fastener is a nonpermanent connection.

a.) Calculate the joint constant, C.

b.) Calculate the recommended preload.

c.) Calculate the load factor against yielding (using the proof strength).

d.) Calculate the factor of safety against joint separation.

e.) Calculate the factor of safety against fatigue failure using the modified Goodman line.

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a.) Calculate the joint constant, C. Answer: 0.249

b.) Calculate the recommended preload. 12.77 kilopounds

c.) Calculate the load factor against yielding (using the proof strength). 4.28

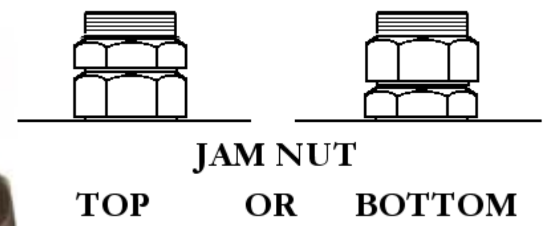
d.) Calculate the factor of safety against joint separation. 4.25

e.) Calculate the factor of safety against fatigue failure using the modified Goodman line. 2.29

Ways to keep Bolts from coming loose under Vibration

- Use a hi-collar lockwasher

- Use a Nylock nut



- Use a jam nut with the standard nut

- Use Loctite or another brand of thread-locking adhesive

Images: wedo.hillmangroup.com, sheldonbrown.com, epcserver.com, meguiarsonline.com, pcarworkshop.com

Dowel Pins

- Dowel pins are used to ensure .001” alignment between mating parts.
- They should be located as far apart as possible, to achieve the best alignment.
- Tolerance on the locations of the holes should be +/-.001” or tighter
- Size of holes for dowel pins: **use a Press Fit (interference fit) into one part, Slip Fit (clearance) into the other**. The slip fit should be large enough to accommodate the range of location tolerance.
- Dowel pins are case-hardened to a depth of about .030”
- Use “pull dowels” on all blind holes. These have a threaded hole in the center for easy removal of the dowel.
- Dowel pins can also be used for withstanding a **shear load** (in combination with threaded fasteners, for withstanding the axial load)

Images: forums.clubrsx.com, almabolt.com

How would you program a single nutrunner to achieve a target torque of 20 Newton-meters +/- 2 Newton-meters?

The nutrunner contains the following components:

- DC electric motor with gearbox
- Encoder
- Torque Transducer
- Socket for engaging with the nut

HINT: Think about starting a screw into a threaded hole and tightening it by hand.
How much torque do you feel with your fingers in each step of the process?
How does that relate to the speed at which you should turn the screw?

Images: directindustry.com, info.admet.com, ferret.com.au